

EDUARDO AMORIM

Senior AI/ML Engineer

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[LinkedIn](#) | [GitHub](#)

SUMMARY

Senior AI/ML Engineer with 6+ years of experience specialized in MLOps, Generative AI, Multi-Agent Systems, and AI Security. Expert in building agentic workflows (LangGraph/MCP) and high-concurrency Task Queue architectures using Python (Celery/Redis). Proven track record in operationalizing LLMs at scale on GCP, integrating observability (LangSmith/Arize), and implementing robust IaC standards (Terraform/Pulumi).

WORK EXPERIENCE

TELUS Digital <i>Senior AI Engineer</i>	Vancouver, Canada (Remote) <i>Aug 2025 – Present</i>
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- Architected end-to-end **Multi-Agent Systems** and AI systems on GCP for 10,000+ users using **LangGraph**, **LangChain**, and **MCP servers**, applying agentic design patterns to query multi-table data warehouses.
- Operationalized and scaled Data Science pipelines by migrating notebook-based research into production-grade workflows using **Kubeflow** and **Vertex AI Pipelines**, implementing telemetry and granular cost monitoring.
- Integrated observability via **LangSmith** and **Arize Phoenix**, reducing costs by 30%, increasing task capacity by 5x, and cutting analysis time from 3 days to seconds.
- Engineered full-stack components (**React** frontends, **Express** BFFs) and RAG workflows with **pgvector** on **Cloud SQL** and **Turbopuffer**, managing IaC with **Terraform** and **Pulumi**.
- Implemented high-scale audio processing features (diarization, silence extraction) with **Celery** and **Redis**, parallelizing intensive pipelines to cut execution time by 46%.
- Orchestrated CI/CD pipelines using **Cloud Build**, **Cloud Deploy**, and **Cloud Run**, while managing sensitive credentials with **Secret Manager** for a secure environment.

Neurotech <i>ML Engineer</i>	Brazil <i>Feb 2023 – Jul 2025</i>
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- Applied hybrid ML algorithms that combined performance and cost-efficiency, resulting in a sale worth **9x the annual corporate target** for a Top-10 Brazilian bank.
- Led the transition of legacy scripts into production-ready Python packages, implementing unit testing and automated documentation.
- Acted as a technical lead in meetings with financial auditors to explain model interpretability and feature engineering strategies.

Voxar Labs <i>ML Engineer</i>	Brazil <i>Aug 2022 – Feb 2023</i>
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- Conducted SOTA research in **Federated Learning** using Flower and PySyft frameworks, resulting in an international publication.
- Deployed drone-based anomaly detection models on **Azure (AKS)** using Docker and Kubernetes, integrated with Grafana for real-time monitoring.

AiBox Lab <i>ML Engineer</i>	Brazil <i>Dec 2020 – Feb 2022</i>
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- Developed an NLP system to generate textual analysis metrics (readability, coherence, and cohesion) using **NLTK** and **SpaCy**.
- Engineered a feedback classifier for university assessments using **Scikit-learn**, achieving **95.1% accuracy** in automating large-scale individual teacher evaluations.

PUBLICATIONS

- **SBSC 2026:** "Operationalizing the Traffic Light Protocol for Native Portuguese Language Model Guardrails in Collaborative Workflows" (AI Security & Collaborative Systems).
- **IJCNN 2026:** "Robustness of Language Models against Portuguese Harmful Prompts" (AI Security & NLP).
- **VISAPP 2023:** "FedBID and FedDocs: A Dataset and System for Federated Document Analysis" (Federated Learning & Computer Vision).

EDUCATION

Centro de Informática UFPE (CIn)
Bachelor of Computer Science | **GPA: 9.1/10.0**

Recife, Brazil
Graduated 2025

TECHNICAL SKILLS

- **Backend & ML:** Python, FastAPI, Celery, Redis, LangChain, LangGraph, MCP, LLMs, PyTorch, Scikit-Learn, OAuth2.
- **GCP Stack:** Cloud Run, Pub/Sub, Firestore, Cloud Build, Cloud Deploy, Cloud SQL (pgvector), Secret Manager, Vertex AI, KubeFlow.
- **Frontend & DevOps:** React, Express, JavaScript, Docker, Kubernetes, Terraform, Pulumi, LangSmith, Arize Phoenix, Turbopuffer, CI/CD, Git.